

Daeen Kabir

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Education

KAIST, Korea Advanced Institute of Science and Technology

Aug 2020 - Aug 2025

B.S. School of Computing (Major)

School of Electrical Engineering (Minor)

Semi-minor in Artificial Intelligence

Work Experience

AI Engineer Intern, MolpaxBio 🔗

Dec 2023 - June 2024

- Developed models for cancer prognosis and staging classification using hospital whole-slide imaging data, achieving accuracy and AUC consistently above 90%.
- Made YOLO-based detection model for liver cellular structures in whole slide images with diagnostic precision of 83% and mIoU of 78%.
- Generated realistic lung tissue whole-slide images using Generative Adversarial Networks (GANs), achieving an FID score of 12.97.
- Applied advanced deep learning techniques, including multi-instance learning, self-supervised learning, and feature-bagging, to enhance diagnostic precision and image generation quality.

Undergraduate Researcher, AMI Lab, KAIST 🔗

Mar 2025 - Present

- Working on optimization-free 3D mesh texturing leveraging correspondence markers.
- Successfully reproduced a diffusion-based novel-view synthesis pipeline that utilizes multi-view epipolar attention without any model training or fine-tuning.

Undergraduate Researcher, SGVR Lab, KAIST 🔗

Jul 2024 - Jun 2025

- Led research on 3D segmentation in dynamic scenes using Gaussian Splatting, novel-view synthesis, and pose-free reconstruction.
- Developed a 4D Gaussian-based interactive segmentation method leveraging motion cues and low-dimensional DINO features for click-based 3D segmentation (**URP Research**).
- Reproduced the training code for a SAM-guided contrastive learning based semantic segmentation of dynamic Gaussians.

Undergraduate Researcher, Users & Information Lab, KAIST 🔗

Sep 2024 - Feb 2025

- Researched LLM's understanding on low-resource language (Bengali).
- Developed a culturally aware dataset for Bengali focusing on historical, cultural, constitutional and linguistics aspects.

Awards

National Scholarship Tier 1

Fall 2020 - Fall 2024

Tuition fee and full amount of school support fees supported by Korea Government

Undergraduate Research Participation (URP) Program

Fall 2024

Among selected recipients to receive *KRW 3 million* funding to engage in research activities

Projects

Progressive Self-Knowledge Distillation (Code)(Paper)

Spring 2023

Enhanced image classification on CIFAR-100 using self-knowledge distillation with mutual learning, improving accuracy while reducing computation.

DINO-Matching 3D Gaussian Splatting (Code)(Paper)

Fall 2023

Improved small object rendering and reduced artifacts by refining scene reconstruction and viewpoint consistency in DINO-feature space.

OECT: Optimized-Embedded Cluster & Tune (Code)(Paper)

Fall 2023

Enhanced LLM classification by integrating an embedding-clustering algorithm, achieving a 2.6% and 4% accuracy boost on topical/non-topical datasets.

Skills & Languages

Languages: Python (5 years), C, Java (2 years), SQL (1 year)

Libraries & Tools: PyTorch, OpenCV, Scikit-Learn, NumPy, Pandas, Seaborn, Apache Spark, Figma, Git, Docker, L^AT_EX, CUDA

Languages: Bengali (*Native*) English (*Full Proficiency*) Korean (*Beginner*)